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# Outcomes of Iodine-125 Plaque Brachytherapy for Uveal Melanoma With Intraoperative Ultrasonography and Supplemental Transpupillary Thermotherapy

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**Summary**

This study reports a retrospective analysis of 526 patients with uveal melanoma treated with iodine-125 (I-125) episcleral plaque brachytherapy with ultrasound-guided plaque localization. Supplemental transpupillary thermotherapy (TTT) was performed in 72 patients (13.7%) at high risk for local recurrence (LR). After median follow-up of 45.9 months, LR was detected in 19 patients (3.6%). Detection of plaque tilt by ultrasonography allows TTT to be used in patients at high

**Purpose:** To assess the impact on local tumor control of intraoperative ultrasonographic plaque visualization and selective application of transpupillary thermotherapy (TTT) in the treatment of posterior uveal melanoma with iodine-125 (I-125) episcleral plaque brachytherapy (EPB).

**Methods and Materials:** Retrospective analysis of 526 patients treated with I-125 EPB for posterior uveal melanoma. Clinical features, dosimetric parameters, TTT treatments, and local tumor control outcomes were recorded. Statistical analysis was performed using Cox proportional hazards and Kaplan-Meier life table method.

**Results:** The study included 270 men (51%) and 256 women (49%), with a median age of 63 years (mean, 62 years; range, 16-91 years). Median dose to the tumor apex was 94.4 Gy (mean, 97.8; range, 43.9-183.9) and to the tumor base was 257.9 Gy (mean, 275.6; range, 124.2-729.8). Plaque tilt >1 mm away from the sclera at plaque removal was detected in 142 cases (27%). Supplemental TTT was performed in 72 patients (13.7%). One or 2 TTT sessions were required in 71 TTT cases (98.6%). After a median follow-up of 45.9 months (mean, 53.4 months; range, 6-175 months), local tumor recurrence was detected in 19 patients (3.6%). Local tumor recurrence was associated with lower dose to the tumor base ($P = .02$).

**Conclusions:** Ultrasound-guided plaque localization of I-125 EPB is associated with excellent local tumor control. Detection of plaque tilt by ultrasonography at plaque removal allows supplemental TTT to be used in patients at potentially higher risk for local recurrence while sparing the majority of patients who are at low risk. Most patients require only 1 or 2 TTT sessions. © 2014 Elsevier Inc.

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risk for LR, and results in excellent local control.

## Introduction

Uveal melanoma is the most common primary intraocular tumor, occurring in approximately 2000 patients per year in the United States (1). Treatment options include enucleation (eye removal) and globe sparing techniques such as episcleral plaque brachytherapy (EPB), charged particle therapy, and stereotactic radiotherapy. Iodine-125 (I-125) EPB is the most common treatment for uveal melanoma in the United States and was the subject of the Collaborative Ocular Melanoma Study (COMS) medium tumor trial, which found no significant difference in disease-specific and overall survival rates for uveal melanoma patients treated with I-125 EPB compared to enucleation (2). Because some studies have reported that local tumor recurrence after EPB is a risk factor for metastasis (3-5), it is important to maximize local tumor control. The COMS reported a local tumor control rate of 89.7% at 5 years (4). However, COMS excluded uveal melanomas located predominantly within the ciliary body, near the optic disc, and those with extraocular tumor extension, which account for a large proportion of uveal melanomas (2). A recent review of the literature revealed a broad range of local tumor control rates with I-125 plaque brachytherapy, with a weighted average local control rate of 90.4%, and with some centers reporting control rates as low as 73% (6).

In our center, several steps are taken to maximize local control. We perform intraoperative ultrasonography at the time of plaque placement to verify correct positioning of the plaque, and at the time of plaque removal to detect plaque tilt that may have occurred during treatment, which is associated with local recurrence (7). Thin, posterior tumors are at higher risk for underdosing and plaque tilt, respectively (7). Consequently, we add 0.5 to 1.0 mm to the prescription point in these selected cases. Finally, we perform supplemental transpupillary thermotherapy (TTT) in selected patients who are at higher risk for local failure (9). In this study, we assess the effectiveness of these measures on local tumor control in patients with posterior uveal melanoma treated with I-125 EPB from 1996 to 2011.

## Methods and Materials

This study was approved by the Washington University Institutional Review Board and comprised 526 consecutive patients who underwent I-125 EPB for posterior uveal melanoma (defined as melanomas arising from the ciliary body and/or choroid) in which intraoperative ultrasonography was performed and at least 6 months' follow-up was available. All surgeries were performed by 1 author (J.W.H).

Baseline clinical characteristics were recorded, including patient age, sex, tumor dimensions and location, and local tumor status at last patient encounter recorded as local tumor control, intraocular recurrence or orbital recurrence. Intraocular tumor recurrence was defined as an increase in tumor thickness or basal dimension >0.5 mm, as assessed by clinical and ultrasound examinations, that occurred beyond 6 months after EPB (such increases in the first 6 months are usually due to tumor edema).

Dosimetric parameters were calculated and recorded as follows. A fundus diagram was created for each patient to transfer anatomical information into a 2-dimensional model eye diagram, from which the coordinates to the critical points were determined. Dosimetry was performed based on COMS proposed formalism and using TG-43 dosimetric algorithms, without accounting for anisotropy or heterogeneity (9,10). A COMS-style gold plaque was used in all cases (2). Plaques were designed to deliver a dose of 85 Gy in 96 hours to the prescription point at a dose rate of 0.8 to 0.9 Gy/h. The prescription point was the tumor apex for tumors >5 mm in thickness, consistent with current guidelines (11).

Patients with tumors less than 5 mm in height were deemed to be at high risk for plaque tilting and were given additional dosimetric allowances (ie, prescription point set to 0.5 to 1 mm above tumor apex), as previously described (7). Plaques were designed with a notch to accommodate the optic nerve when tumors were located within 1.5 mm of the optic disc. The techniques for plaque insertion and removal, and for intraoperative ultrasonography were previously described (7). Briefly, at least 3 and as many as five 5-0 nylon sutures were used to secure the plaque to the sclera. Intraoperative B-scan mode ultrasonography was performed under sterile conditions at plaque insertion to ensure optimal localization and to adjust for plaque tilting away from the sclera. Beginning in 2000, intraoperative ultrasonography was performed again at plaque removal to assess whether plaque tilting had occurred since the time of plaque placement, as previously described (7).

Supplemental transpupillary thermotherapy (TTT; also called diode laser hyperthermia) was performed in selected cases using an 810-nm wavelength diode laser delivered with a large spot size (0.5-3.0 mm) and long duration (1 minute per spot), as previously described (8). The primary indication for adjuvant TTT was tilt >1 mm, which was more likely for juxtapapillary tumors (<1.5 mm from the optic disc), but also occurred for some other posterior pole tumors. Supplemental TTT was also performed when there was a delayed reduction in exudative retinal detachment and/or tumor thickness at 3-6 months after EPB. In this situation, TTT was continued at 3- to 4-month intervals until tumor thickness decreased and retinal detachment had resolved. TTT was also performed in selected patients with suspected local tumor recurrence, in which case TTT was continued at 3- to 4-month intervals until tumor control was achieved.

Freedom from local progression was calculated using the Kaplan-Meier method. The Cox proportional hazards model was used for univariate analysis. Significance was defined as a $P$ value $\leq$.05. Statistical analyses were performed using SAS, version 9.2 (SAS Institute, Cary, NC).

## Results

### Clinical features

Table 1 summarizes the clinical features of the 526 patients in this study. Median age was 62.8 years (mean, 62.2 years; range, 16-91 years). Sex was male in 270 patients (51%) and female in 256 patients (49%). Median tumor thickness was 4.2 mm (mean, 5.0 mm; range, 1.4-14.3 mm). Median largest basal tumor dimension

**Table 1** Clinical characteristics of study patients

| Characteristic | n | % |
|---|---|---|
| Age at plaque radiotherapy, y | | |
| 16-39 | 32 | 6 |
| 40-59 | 187 | 36 |
| 60-79 | 252 | 48 |
| >80 | 55 | 10 |
| Sex | | |
| Male | 270 | 51 |
| Female | 256 | 49 |
| Largest basal tumor dimension (mm) | | |
| <8.0 | 65 | 12 |
| 8.0-16 | 390 | 74 |
| >16.0 | 71 | 13 |
| Tumor thickness (mm) | | |
| <3.0 | 93 | 18 |
| 3.0-8.0 | 366 | 70 |
| >8.0 | 67 | 13 |
| Ciliary body involvement | 96 | 18 |
| Juxtapapillary location (≤1.5 mm from disc) | 144 | 27 |
| Pathologic cell type* | | |
| Spindle | 56 | 30 |
| Mixed | 69 | 37 |
| Epitheliod | 60 | 32 |
| TNM clinical stage | | |
| T1 | 130 | 25 |
| T2 | 228 | 43 |
| T3 | 157 | 30 |
| T4 | 12 | 2 |
| Not eligible for COMS study | 221 | 42 |
| Follow-up (all patients) | | |
| Median/mean (range), mo | 45.9/53.4 (6-175) | |
| ≥1 y | 480 | 91 |
| ≥2 y | 389 | 74 |
| ≥3 y | 308 | 59 |
| ≥4 y | 252 | 48 |
| ≥5 y | 184 | 35 |

*Abbreviation:* COMS = Collaborative Ocular Melanoma Study.
American Joint Committee on Cancer (AJCC) 7th edition
* Data available for 185 patients.

was 12 mm (mean, 12.2 mm; range, 3-22 mm). Ciliary body involvement was present in 95 patients (18%). Median distances to the disc and fovea were 4.0 mm and 4.0 mm, respectively. The American Joint Committee on Cancer (AJCC) seventh edition clinical categories were T1 in 130 patients (25%), T2 in 228 patients (43%), T3 in 156 patients (30%), and T4 in 12 patients (2%). The COMS medium tumor trial would have excluded 221 patients (42%) because of 1 or more of the following factors: large tumor size in 97 patients, juxtapapillary tumor location in 144 patients, predominantly ciliary body location in 51 patients, and extraocular extension in 2 patients.

## Episcleral plaque brachytherapy

Table 2 summarizes the I-125 EPB treatment characteristics. Mean treatment depth was 5.7 mm (median, 5 mm; range, 2.8-13 mm). The amount of plaque tilt at removal was recorded in 432 patients. It was nil in 191 patients (44.2%), ≤1 mm in 99 patients (22.9%), and >1 mm in 142 patients (32.9%). Dose to the prescription point was calculated to the actual tumor thickness in 167 cases (31.8%), whereas in 342 cases (65%) the prescription point was a median of 1 mm (mean, 1 mm; range, 0.1-3 mm) greater than the tumor thickness because the tumors were at high risk for local failure. The prescription point height was greater than tumor thickness in 110 (76.4%) patients with juxtapapillary tumors. The mean dose to the prescription point was 85.3 Gy (median, 85.3 Gy; range, 47.5-150.6 Gy). The mean doses to the tumor apex and base were 97.8 Gy (median, 94.4 Gy; range, 43.9-183.9 Gy) and 275.6 Gy (median 257.9 Gy; range, 124.2-729.8), respectively.

**Table 2** Summary of iodiine-125 (I-125) episcleral plaque brachytherapy parameters

| Characteristic | n | % |
|---|---|---|
| Plaque size (mm) | | |
| 12-14 | 126 | 24 |
| 16-18 | 222 | 42 |
| 20-23 | 178 | 34 |
| Notched plaque | 123 | 23 |
| Plaque tilt at removal (mm)* | | |
| 0 | 191 | 44 |
| ≤1 | 99 | 23 |
| >1 | 142 | 33 |
| Dose to tumor base (Gy)† | | |
| ≤230 | 194 | 37 |
| >230-350 | 229 | 44 |
| >350 | 100 | 19 |
| Dose to tumor apex (Gy) | | |
| ≤85 | 107 | 20 |
| >85-115 | 359 | 68 |
| >115 | 60 | 11 |
| Increased dose to tumor apex due to high risk features | 279 | 53 |

* Data available on 432 patients.
† Data available on 523 patients.

Supplemental TTT was performed in 72 patients (13.7%). The numbers of TTT sessions were 1 in 63 patients (87.5%), 2 in 8 patients (11.1%), and 3 in 1 patient (1.4%). Supplemental TTT was performed in 21 patients (29.2%) because of juxtapapillary location or plaque tilting, in 6 patients (8.3%) because of frank local recurrence, and in 45 patients (62.5%) because of slow tumor regression or exudative retinal detachment.

## Local tumor control

Local tumor control outcomes are summarized in Fig. 1. After a median follow-up of 45.9 months (mean, 53.4 months; range, 6-175 months), local tumor recurrence was detected in 19 patients (3.6%), including 15 intraocular recurrences (2.9%) and 4 orbital recurrences (0.7%). Median time to local tumor recurrence was 23.6 months. Of the 19 patients with local recurrence, 18 were deemed to have local recurrence because of an increase in tumor diameter, whereas the remaining patient was judged to have a local recurrence because of an increase in tumor thickness.

Of the 4 patients who developed orbital recurrence, 3 presented as atypical cases that would have been excluded from the COMS. The first of these 3 atypical patients had a very large tumor with

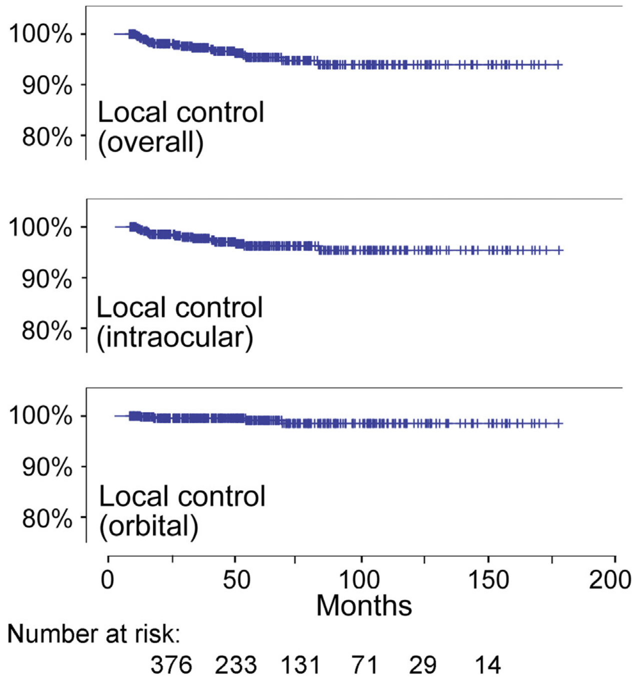


**Fig. 1.** Kaplan-Meier plots demonstrating local tumor control after iodine-125 (I-125) plaque brachytherapy. Overall, intraocular and orbital recurrences are shown, as indicated, with median follow-up of 45.9 months (mean, 53.4 months; range, 6-175 months).

largest basal dimension of 22 mm. Enucleation was recommended, but the patient refused and was then lost to follow-up for 3 years after EPB. This patient eventually required exenteration and later died of metastatic disease. The second patient had a large extraocular tumor component at initial diagnosis that was excised before insertion of the plaque. The third atypical patient who developed orbital recurrence had a small area of extraocular tumor extension detected by ultrasound examination before EPB, which was accounted for in plaque dosimetry but which would have disqualified the patient for the COMS.

Treatment of intraocular recurrence was TTT alone in 2 patients and enucleation in 13 patients. Treatment of orbital recurrence was surgical excision with repeat EPB in 2 patients, exenteration in 1 patient, and observation in 1 patient who also had advanced metastatic disease at the time of local recurrence.

The estimated 3- and 5-year actuarial local control rates were 97.3% and 95.4%, respectively. The estimated 3- and 5-year actuarial local control rates for patients with juxtapapillary tumors were 96.8% and 95.6%, respectively. The only clinical factor associated with local tumor recurrence was lower radiation dose to the tumor base ($P = .02$). There was a non-significant trend toward association between local recurrence and larger basal tumor diameter, increased TNM clinical category and plaque tilt at removal (Table 3).

**Table 3** Cox proportional hazards analysis of clinical factors and local tumor recurrence

| Clinical feature | Univariate *P* value |
|---|---|
| Patient age | .80 |
| Sex | .25 |
| Largest basal tumor dimension | .08 |
| Tumor thickness | .24 |
| Ciliary body involvement | .94 |
| Pathologic cell type | .51 |
| Juxtapapillary location | .60 |
| TNM clinical category | .10 |
| Patients not eligible for COMS | .99 |
| Plaque size | .31 |
| Notched plaque | .60 |
| Plaque tilt at removal | .11 |
| Dose to tumor base | .02 |
| Dose to tumor apex | .95 |
| Prescription depth greater than tumor thickness due to high-risk features | .56 |

## Discussion

I-125 EPB is the most common treatment for posterior uveal melanoma in the U.S. and many other countries (1). Yet, the techniques for performing this procedure, the use of intraoperative ultrasonography and supplemental TTT, and the reported local control rates vary widely among centers (6). This study shows that the technique that we have developed results in local tumor control rates that are comparable or superior to EPB and proton beam radiotherapy results from other centers reporting similar patient numbers and follow-up periods. The median long basal tumor dimension in our study was 12 mm, and tumor thickness was 4.2, comparable to the weighted averages of 11.1 and 4.8 mm, respectively, reported by other institutions treating uveal melanoma with I-125 EPB, and slightly smaller than the weighted averages of 14.0 and 5.5 mm, respectively, reported by institutions treating with proton beam therapy (6). In particular, our local control was higher than the 90.4% weighted average rate reported by other institutions treating with I-125 EPB, and much higher than that reported for the COMS despite our inclusion of patients who would not have been eligible for the COMS because of tumor size, juxtapapillary location, or extraocular tumor extension, which are risk factors for local recurrence (4).

Key features of our approach include the following: (*1*) intraoperative ultrasonography at plaque insertion to ensure optimal localization and at plaque removal to detect plaque tilting; (*2*) selective application of supplemental TTT in patients at high risk for local recurrence; and (*3*) increased radiation prescription dose to the tumor apex for posteriorly located tumors <5 mm thick.

Centers that use intraoperative ultrasonography for EPB consistently report high local control rates (7,12,13). Notably, the COMS did not require intraoperative ultrasonography, which may partly explain the lower local control rate in this study. Ultrasonography at plaque insertion allows the plaque to be repositioned as needed to optimize its localization over the tumor. In addition, plaque tilting, which cannot be detected without ultrasonography, can be corrected by various surgical techniques such as readjusting the plaque, dis-inserting an extraocular muscle, or securing the plaque to the sclera with a mattress suture. Despite such efforts, however, plaque tilting can occur during the time that the plaque is in place, because of periocular tissue swelling or episcleral hematoma, and this can be detected with ultrasonography at plaque removal (7).

Supplemental TTT also plays an important role in our approach to EPB. This treatment was selectively used in patients with high risk features, namely, for juxtapapillary tumor location or for plaque tilt >1 mm at plaque removal. This selective use of TTT allowed us to safely withhold this treatment in the vast majority of patients (∼85%) while successfully achieving local control in most high-risk patients. Furthermore, we found that, within the given follow-up, only 1 session of TTT was sufficient in the vast majority of patients, with only 12.5% of patients who received TTT needing more than 1 session. In contrast to our technique of using TTT selectively after EPB for patients with high-risk features, Yarovoy et al reported their use of TTT simultaneously with EPB for patients with large tumors. They described higher rates of local control and eye preservation in patients receiving TTT in addition to EPB, compared with patients receiving EPB alone, while maintaining similar complication rates and visual outcomes (14). We are currently analyzing the radiation complication rates in our patients who received TTT after EPB to determine whether they differ significantly from those in patients who received EPB alone.

In the COMS, tumors <5 mm in thickness were nevertheless treated to a prescription dose of 5 mm, which led to apical treatment doses much greater than 85 Gy and greater risk of radiation complications in many cases. On the other hand, the risk of local failure due to plaque tilt is greater for thin posterior tumors (7). To address these issues, we have used a protocol in which increased apical treatment doses are reserved for thin posterior tumors. In particular, tumors <5 mm in thickness and located within 3 mm of the fovea or optic disc are treated to a prescription point 0.5 to 1.0 mm beyond the tumor apex. Our results indicate that this modification has had no detectable negative impact on local tumor control. We are currently analyzing the radiation complication rates in this patient cohort to determine how our approach compares to that of the COMS.

Limitations of this study include a retrospective noncomparative study design, limited follow-up, and potential underreporting of local failures because of loss of patients to follow-up. Nevertheless, our findings favor the continued use of the COMS proposed formalism, and strongly support the use of intraoperative ultrasonography and selective use of supplemental TTT in the treatment of posterior uveal melanoma with I-125 EPB.

## Conclusions

In conclusion, ultrasound-guided plaque localization of I-125 EPB is associated with excellent local tumor control. Detection of plaque tilt by ultrasonography at plaque removal allows supplemental TTT to be targeted to patients at potentially higher risk for local recurrence, while sparing the majority of patients who are at low risk. Most patients that receive TTT require only 1 or 2 sessions.

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